

Confounders and controls

Project: Reproducible AI Research Study

1. Your mission

Your learning target

I can explain the mechanism and terminology of confounders and controls.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of confounders and controls

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Experimental units and independence. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My starting explanation

This week's end goal

Explain confounders and controls using the records and boundaries of this fictional investigation.

2. Notice the evidence

A confounder changes alongside the intended factor and can offer another explanation. Controls aim to isolate the proposed mechanism.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

List the given facts and one fact that is not supplied.

3. See how it works

For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.

Confounders and controls — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: New method uses better corpus and different queries while old method uses the original set.

Check or calculation: For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.

Conclusion and limit: Corpus and case differences confound the method effect. Rerun matched inputs or report a whole-system comparison without isolation.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

New method uses better corpus and different queries while old method uses the original set. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: New cases still use only healthy-white and spotted-blue.

Prediction: The collection does not resolve the background confounding.

Reason: The same two variables remain coupled instead of being varied separately.

5. Words that unlock the idea

Word · Meaning

Word: Confounder

Meaning: A differing condition offering another explanation

Word: Control

Meaning: A design constraint preserving comparison conditions

Word: Counterbalancing

Meaning: Varying order to reduce systematic order differences

Use two terms to explain a step in this week's example.

6. Follow a worked example

New method uses better corpus and different queries while old method uses the original set.

Corpus and case differences confound the method effect. Rerun matched inputs or report a whole-system comparison without isolation.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Confounders and controls

Separate background from class. Training data show every healthy leaf on white paper and every spotted leaf on blue paper.

The facts we will use

Goal is leaf condition, not paper colour. A proposed check uses both paper colours for each condition.

Cross class with paper background

	White paper	Blue paper
Healthy	Collect	Collect
Spotted	Collect	Collect

Read a cell using both its row and column labels.

Confounders and controls

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify coupled variables

Condition and background change together in the original collection.



2. Cross the conditions

Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.



3. Inspect the shortcut

Compare predictions when the same leaf's background changes under a controlled test.

CHANGE ONE THING

New cases still use only healthy-white and spotted-blue.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Identify coupled variables

What happens and why: Condition and background change together in the original collection.

Step: 2. Cross the conditions

What happens and why: Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.

Step: 3. Inspect the shortcut

What happens and why: Compare predictions when the same leaf's background changes under a controlled test.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A confounder can provide an alternative explanation for a result. Crossing conditions makes the intended feature and the nuisance feature less entangled.

What this example does not establish

A changed-background test can diagnose sensitivity; it does not prove the model's complete internal reasoning.

Read the labelled picture: Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?

Change one thing: New cases still use only healthy-white and spotted-blue. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

Controls address specified alternatives, not every possible unknown confounder.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Controls address specified alternatives, not every possible unknown confounder. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Confounders and controls”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Confounders and controls. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

New method uses better corpus and different queries while old method uses the original set. Show the intermediate steps and give the conclusion with its scope.

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: Controls address specified alternatives, not every possible unknown confounder. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled confounders and controls evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of confounders and controls. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week’s record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Confounders and controls — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: New method uses better corpus and different queries while old method uses the original set.

Check or calculation: For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.

Conclusion and limit: Corpus and case differences confound the method effect. Rerun matched inputs or report a whole-system comparison without isolation.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Apply a new case independently

Every method A run is morning and B run afternoon; network speed varies by time. Propose a repair.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Every method A run is morning and B run afternoon; network speed varies by time. Propose a repair.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Separate background from class. Training data show every healthy leaf on white paper and every spotted leaf on blue paper.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of confounders and controls

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of confounders and controls

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Separate background from class. Training data show every healthy leaf on white paper and every spotted leaf on blue paper.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Compare predictions when the same leaf's background changes under a controlled test.

1. Identify coupled variables
2. Inspect the shortcut
3. Cross the conditions

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Inspect the shortcut Compare predictions when the same leaf's background changes under a controlled test. A confounder can provide an alternative explanation for a result. Crossing conditions makes the intended feature and the nuisance feature less entangled.

Practice 2: explain the deciding evidence

Changed-case practice: New cases still use only healthy-white and spotted-blue. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. The collection does not resolve the background confounding.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The collection does not resolve the background confounding. The same two variables remain coupled instead of being varied separately.

Repair a missing building block

Review Experimental units and independence (ai-7/9). Goal: Explain the mechanism and terminology of experimental units and independence. Short practice: Count independent plants and photo views. Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the mechanism and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

Review Methods results and limitations (ai-7/4). Goal: Explain the mechanism and terminology of methods results and limitations. Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor evidence in their proper categories. Why cannot the result establish that B is best for outdoor farming?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.